

INTERNET TECHNOLOGY & WEB PAGE DESIGN

Unit-2: World Wide Web

Comprehensive Study Notes

HPU BCA Semester IV

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1. World Wide Web

The **World Wide Web (WWW)** is a system of interlinked hypertext documents and resources that are accessed via the Internet using web browsers.

"The World Wide Web is an information system where documents and other web resources are identified by URLs, interlinked by hypertext, and can be accessed via the Internet."

1.1 Key Features of WWW

Feature	Description
Hypertext	Text with links to other documents
Hypermedia	Hypertext with multimedia elements
Client-Server Model	Users request, servers deliver

Global Accessibility	Accessible from anywhere
Platform Independent	Works on any operating system
User-Friendly	Easy to navigate and use

1.2 URL Structure

A **URL (Uniform Resource Locator)** consists of:

```
protocol://domain:port/path?query#fragment  
https://www.example.com:443/path/page.html?key=value#section
```

Component	Description	Example
Protocol	Communication protocol	https://
Domain	Website address	www.example.com
Port	Connection port (optional)	:443
Path	Resource location	/path/page.html
Query	Query parameters	?key=value
Fragment	Section within page	#section

2. Web Browsers

A **Web Browser** is a software application used to access and display web pages and other resources on the World Wide Web.

2.1 Popular Web Browsers

Browser	Developer	Engine
Google Chrome	Google	Blink
Mozilla Firefox	Mozilla	Gecko
Microsoft Edge	Microsoft	Blink
Safari	Apple	WebKit
Opera	Opera Software	Blink

2.2 Browser Functions

Function	Description
Rendering	Display web pages correctly
Navigation	Go forward, backward, refresh
Bookmarking	Save favorite pages
History	Track visited pages
Security	SSL/TLS, pop-up blocking
Extensions	Add additional functionality
Developer Tools	Inspect and debug web pages

2.3 Browser Security Features

Feature	Description
SSL/TLS	Encrypts communication

Pop-up Blocker	Blocks unwanted pop-ups
Phishing Protection	Detects fraudulent websites
Sandboxing	Isolates web pages
Private Browsing	No history or cookies saved

2.4 HTTP & HTTPS

Feature	HTTP	HTTPS
Full Form	Hypertext Transfer Protocol	Hypertext Transfer Protocol Secure
Encryption	No	Yes (SSL/TLS)
Default Port	80	443
Security	Less secure	Secure
Use	Basic browsing	Banking, transactions

2.5 HTTP Methods

Method	Description
GET	Retrieve data from a server
POST	Send data to a server to create a resource
PUT	Update or replace an existing resource
DELETE	Remove a resource from the server
PATCH	Partially modify a resource
HEAD	Retrieve headers without body

2.6 HTTP Status Codes

Code	Category	Description
1xx	Informational	Request received, continuing
2xx	Success	Request successfully processed

3xx	Redirection	Further action needed
4xx	Client Error	Request contains error
5xx	Server Error	Server failed to process request

Common Codes: 200 OK, 201 Created, 301 Moved Permanently, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable.

2.7 Cookies

Cookies are small text files stored on a user's computer by a web browser to track and remember information.

Type	Description
Session Cookies	Deleted when browser is closed
Persistent Cookies	Remain for a set duration
First-Party Cookies	Set by the visited website
Third-Party Cookies	Set by external domains (ads, analytics)

2.8 Search Engines

Search Engines are software systems designed to search for information on the World Wide Web.

Search Engine	Developer	Type
Google	Google LLC	Crawler-based
Bing	Microsoft	Crawler-based
Yahoo	Yahoo Inc.	Hybrid
DuckDuckGo	DuckDuckGo	Crawler-based (private)

Working of Search Engines

1. **Crawling:** Bots (spiders) discover and fetch web pages
2. **Indexing:** Parse and store page content in a database
3. **Ranking:** Rank results by relevance using algorithms

Types of Search Engines

Type	Description	Example
Crawler-Based	Use automated bots to index pages	Google, Bing
Human-Powered (Directories)	Curated by human editors	DMOZ (archived)
Hybrid	Combination of crawler and human	Yahoo

3. Web Servers

A **Web Server** is software that delivers web pages and other content to clients over the Internet.

3.1 Popular Web Servers

Server	Developer	Features
Apache	Apache Foundation	Open source, widely used, modular
Nginx	Nginx Inc.	High performance, lightweight, event-driven
Microsoft IIS	Microsoft	Windows integration, .NET support
Tomcat	Apache Foundation	Java servlets and JSP
Node.js	Open Source	JavaScript-based, non-blocking

3.2 Web Server Functions

Function	Description
Content Delivery	Serve web pages and files
Request Handling	Process HTTP requests
Security	Authentication, encryption
Load Balancing	Distribute traffic
Caching	Improve performance
Logging	Record access and errors

4. Internet Connections

4.1 Dial-Up Connection

Dial-Up uses a telephone line to connect to the Internet via a modem.

Feature	Description
Speed	Up to 56 Kbps
Technology	Analog modem over telephone lines
Availability	Widely available
Cost	Low cost
Disadvantage	Slow, phone line unavailable during use

4.2 SLIP and PPP

SLIP (Serial Line Internet Protocol) and **PPP (Point-to-Point Protocol)** are protocols used for dial-up connections.

PPP Feature	Description
Authentication	PAP, CHAP for security
Error Detection	CRC for error checking
Compression	Data compression
Dynamic IP	Automatic IP assignment
Multi-protocol	Supports multiple protocols

4.3 Dedicated Lines

Type	Speed
T1 Line	1.544 Mbps
T3 Line	44.736 Mbps
DSL	1-100 Mbps

Cable	10-1000 Mbps
Fiber Optic	100-10000 Mbps

4.4 ISDN

ISDN (Integrated Services Digital Network) is a digital telephone network that provides high-speed connections.

Feature	Description
Speed	64-128 Kbps
Channels	2 B-channels (64 Kbps each)
Digital	Fully digital transmission
Multiplexing	Multiple services on one line

5. TCP/IP Protocol Suite

TCP/IP (Transmission Control Protocol/Internet Protocol) is a suite of communication protocols used to interconnect network devices on the Internet.

5.1 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Error Checking	Yes	Optional
Ordering	Ordered delivery	No ordering
Speed	Slower	Faster
Use	Web, Email, FTP	Streaming, DNS
Header Size	20-60 bytes	8 bytes

6. Domain Name System (DNS)

DNS (Domain Name System) is the phonebook of the Internet that translates domain names to IP addresses.

6.1 DNS Resolution Process

1. User enters a domain name in the browser
2. Browser checks local cache for IP address
3. Query sent to recursive DNS resolver
4. Resolver queries root name servers
5. Root server directs to TLD name server
6. TLD server directs to authoritative name server
7. Authoritative server returns the IP address
8. Browser connects to the IP address

7. Domain Names

A **Domain Name** is a human-readable address used to identify a website on the Internet.

7.1 Domain Name Structure

subdomain.example.com

| --subdomain-- | --SLD-- | TLD |

Component	Description	Example
TLD (Top-Level Domain)	Highest level in domain hierarchy	.com, .org, .in
SLD (Second-Level Domain)	Main name chosen by registrant	example
Subdomain	Prefix added for organization	www, mail, blog

7.2 Top-Level Domains (TLDs)

TLD	Description
.com	Commercial organizations
.org	Non-profit organizations
.net	Network organizations
.edu	Educational institutions
.gov	Government entities
.mil	Military organizations
.in	India
.uk	United Kingdom
.us	United States

7.3 Domain Registration Process

Step	Description
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1	Choose a domain name
2	Check availability (WHOIS lookup)
3	Register with a registrar
4	Pay registration fees
5	Use the domain
6	Renew periodically

8. IP Addressing

An **IP Address** is a unique numerical identifier assigned to each device on a network.

8.1 IPv4 vs IPv6

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Address Format	192.168.1.1	2001:0db8::1
Number of Addresses	4.3 billion	340 undecillion
Security	Optional (IPsec)	Built-in (IPsec)
Configuration	Manual/DHCP	Auto-configuration

8.2 IP Address Classes

Class	Range	Use
A	1.0.0.0 - 126.255.255.255	Large networks
B	128.0.0.0 - 191.255.255.255	Medium networks
C	192.0.0.0 - 223.255.255.255	Small networks
D	224.0.0.0 - 239.255.255.255	Multicast
E	240.0.0.0 - 255.255.255.255	Experimental

8.3 Subnet Mask

A **subnet mask** is a 32-bit number that divides an IP address into network and host portions. For example, **255.255.255.0** (Class C default) means the first 24 bits identify the network and the last 8 bits identify the host.

8.4 Public vs Private IP

Class	Private IP Range
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A	10.0.0.0 - 10.255.255.255
B	172.16.0.0 - 172.31.255.255
C	192.168.0.0 - 192.168.255.255

Public IP: Globally unique, routable on the Internet.

Private IP: Used within a local network, not routable on the Internet.

9. Routing

Routing is the process of forwarding data packets between networks.

9.1 Routing Protocols

Protocol	Type	Description
RIP	Distance Vector	Simple, small networks
OSPF	Link State	Fast convergence, large networks
BGP	Path Vector	Internet routing

Questions

Short Answer Questions (2-5 Marks)

1. What is WWW?
2. What is a web browser?
3. What is a web server?
4. What is TCP/IP?
5. What is DNS?
6. What is a domain name?
7. What is an IP address?
8. What is the difference between IPv4 and IPv6?
9. What is SLIP?
10. What is PPP?

Long Answer Questions (10 Marks)

1. Explain the architecture of WWW.
2. Explain DNS and domain name registration.
3. Explain TCP/IP protocol suite.
4. Explain IP addressing and its classes.
5. Explain different types of Internet connections.

Objective Questions (1 Mark)

1. WWW stands for:
(a) World Wide Web (b) Wide World Web (c) Web World Wide (d) None
2. DNS stands for:
(a) Domain Name System (b) Data Name System (c) Domain Network System
(d) None
3. IPv4 uses:
(a) 32 bits (b) 64 bits (c) 128 bits (d) None
4. IPv6 uses:

(a) 32 bits (b) 64 bits (c) 128 bits (d) None

5. PPP stands for:

(a) Point-to-Point Protocol (b) Packet-to-Packet Protocol (c) Point Protocol
(d) None